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ASSASSINATING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyse knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
Total Marks:30

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I K. Akshitha (192312527) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Akshitha.

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SP

Smart Device Connectivity.1. Proportional logic.

let

- W = Smart TV is connected to WiFi.
- I = Smart TV can access the internet.

Statements:

1. $W \rightarrow I$

2. W

Goal : I .2. CNF and Negation of Goal.

Implication:

$$W \rightarrow I \equiv \neg W \vee I.$$

CNF

- $\neg W \vee I$
- W
- Negated goal : $\neg I$.

3. Resolution steps:-

Resolve:

$$(\neg W \vee I), W.$$

∴ I , Now resolve: $I, \neg I$.Therefore $\square$ 4. Conclusion

Since the empty clause $\square$ is obtained, the goal is proved.

Smart TV can access the internet.

② Machine Auto shutdown.

1. Proportional logic

let

- O = machine overshoots
- S = machine shuts down automatically.

Statements:

1. $O \rightarrow S$

2. O

Goal: S .

2. CNF and Negation of Goal.

$$O \rightarrow S \equiv \neg O \vee S$$

CNF:

- $\neg O \vee S$
- O
- Negative goal: $\neg S$.

3. Resolution steps

First: $(\neg O \vee S), O$.

Resolve O & $\neg O$ S .

Next S $\therefore S, \neg S$.

$\square$

4. Conclusion

The empty clause is obtained, so the goal is proved.

The machine shuts down automatically.

③ Account Verification.

1. Proportional logic.

let

- V = user uploads a valid ID
- A = Account is verified.

2. CNF & Negation of Goal

$$P \rightarrow A \equiv \neg P \vee A$$

CNF:

- $\neg P \vee A$
- P
- Negative goal: $\neg A$

3. Conclusion

The empty clause is obtained, so the goal is proved.

Anitha gets premium access.

⑤

Vehicle Registration process.

1. Propositional logic.

let:

- E = car passes the emission test
- R = car is certified roadworthy
- G = car can be registered.

2. convert into CNF

State: 1 $E \rightarrow R$
become $\neg E \vee R$

State: 2 $R \rightarrow G$
become $\neg R \vee G$

State 3: E

3. Conclusion

The empty clause $\square$ is obtained. Therefore, the original goal is proved.

The car can be registered.

Since the empty clause is derived, the goal "the car can be registered" is true according to the knowledge base.

Statements :

1. $V \rightarrow A$

2. $A \vee V$

Goal: A .

2. CNF and Negation of Goal

$$V \rightarrow A \equiv \neg V \vee A$$

CNF:

• $\neg V \vee A$

• V

• Negated goal: $\neg A$.

3. Resolution steps

Resolve: $(\neg V \vee A), V$

we get: A

Now resolve: $A, \neg A$

$\therefore \quad \textcircled{Q}$

4. Conclusion

The empty clause is obtained

The account is verified.

④ Premium subscription Access.

1. Propositional logic

let

• P = Anitha paid the subscription fee.

• A = Anitha gets premium access.

Statements :

1. $P \rightarrow A$

2. P

Goal: A